

COS AT-1 Pseudocode
Writing Task

Date _____
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① Prim's for communication Latency

$T \leftarrow \emptyset$, $Total \leftarrow 0$

Select any starting vertex

Add its edges to a priority queue

while queue is not empty:

 select minimum-weight edge (u, v)

 If v is already visited, continue

 If $Total + weight(u, v) > C_{max}$:

 Return T

 Add (u, v) to T

$Total \leftarrow Total + weight(u, v)$

 Mark v as visited

 Add v 's edges to queue

Return T

② Kruskal's Algorithm - Maximum Spanning Tree

$T \leftarrow \emptyset$

Make separate sets for all vertices

Sort edges in decreasing order of weight

For each edge (u, v) :

 If $Find(u) \neq Find(v)$:

 Add (u, v) to T

 union (u, v)

 If $|T| > |V| - 1$:

 Break

Return T